

Advay Kadam

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. Computer Science and Statistics, Minor in Mathematics | GPA: 3.9

May 2027

- **Selected Coursework:** Graduate Distributed Systems, Machine Learning Systems, Operating Systems, Deep Generative Models, Applied Parallel Programming, Computer Architecture, Data Structures in C++
- **Awards:** **Top 3%** at IMC Prosperity (International Trading Challenge), **2x** Dean's List
- **Course Assistant:** Data Structures and Algorithms in C++, Statistical Modeling (Applied Regression)

EXPERIENCE

Amazon

Feb 2026 – Present

Incoming Software Development Engineer Intern

New York City, NY

- Incoming Summer 2026 SDE Intern on Amazon Traffic Engineering, building scalable network infrastructure

Supercomputing Systems and AI Lab @ UIUC

Aug 2025 – Present

Machine Learning Systems Researcher (PI: Dr. Minjia Zhang)

Urbana, IL

- Developed a fused RMSNorm + Linear kernel in AWS Neuron Kernel Interface, reducing operator latency by **10%** over native PyTorch Neuron on Trainium 1 hardware
- Extended the kernel to Trainium 2 with 2-NeuronCore sharding, scaling execution across cores for larger tensor shapes and RL rollout inference
- Integrated the kernel into Qwen3-8B-style rollout generation, improving end-to-end inference latency by **2%** across sequence-length sweeps

Bank of New York Mellon

Jun 2025 – Aug 2025

Software Engineer Intern

Jersey City, NJ

- Built a FastAPI service for a LangGraph/LangChain workflow with **4** agents + orchestrator, generating explainable column-level security labels from dataset metadata
- Integrated AWS SQS to process **500+** daily dataset requests across LOBs, cutting end-to-end latency by **30%** with chunking and bounded concurrency
- Deployed AWS Lambda workers to run asynchronous agent pipelines and persist **30,000+** security labels to S3

Procter & Gamble

May 2024 – Aug 2024

Software Engineer Intern

Cincinnati, OH

- Optimized a sales ETL pipeline in Azure Databricks, rewriting SQL/PySpark transformations to cut end-to-end runtime by **40%** and reduce cluster cost by **25%**

PROJECTS

Hybrid Distributed File System (HyDFS) | C, Sockets (UDP/TCP), Multithreading, Linux

- Built a Cassandra-inspired distributed file system across **10** Linux machines in C, implementing consistent hashing, UDP gossip membership, TCP file transfer, and 3-way replication
- Improved replication and merge protocols for eventual consistency, increasing throughput by **2.7x** and lowering network/synchronization overhead by **35%**

Effective Context Utilization in LLMs | Python, PyTorch, Hugging Face Transformers, NumPy

- Evaluated 6 attribution methods across 3 context lengths and 5 needle depths, profiling peak VRAM, wall-clock latency, and prompt-region influence on an NVIDIA H200 GPU
- Implemented batched attention-mask and substitution ablations, reducing profiling runtime by up to **3.48x** versus physical chunk deletion while preserving RoPE token positions

Fused LayerNorm + GELU Kernel | C++, CUDA

- Designed a fused LayerNorm + GELU CUDA kernel for NVIDIA A100, reducing global memory traffic by combining normalization and activation into a single pass; achieved **1.5x** speedup over PyTorch

TECHNICAL SKILLS

Languages: Python, C, C++, Java, SQL

ML Frameworks + Libraries: PyTorch, CUDA, LangChain/LangGraph, AWS Neuron Kernel Interface, NumPy

Systems + Tools: Linux, Bash, Git, Docker, Networking (TCP/UDP), AWS (SQS, Lambda, EC2), FastAPI, PySpark